Estimation and Hypothesis Testing

Course notes - Department of Mathematics

1. Populations and samples

Statistics infers properties of a population from a sample. The inference is only as good as the sampling: a biased sample cannot be rescued by a larger size, it merely produces a more confident wrong answer.

A statistic is any function of the sample. Its distribution across hypothetical repeated samples - the sampling distribution - is what licenses every inferential claim.

2. Point estimation

An estimator is unbiased when its expectation equals the parameter. Consistency means it converges to the parameter as the sample grows. Efficiency compares variances among unbiased estimators.

Maximum likelihood chooses the parameter making the observed data most probable. It is consistent and asymptotically efficient under regularity conditions, which is why it is the default method.

3. Confidence intervals

A 95 percent confidence interval is constructed so that the procedure captures the true parameter in 95 percent of repeated samples. The probability belongs to the procedure, not to any particular interval already computed.

Interval width shrinks with the square root of the sample size, so halving the width requires quadrupling the data.

4. Hypothesis testing

A test contrasts a null hypothesis with an alternative. The p-value is the probability, assuming the null is true, of observing a statistic at least as extreme as the one obtained. It is not the probability that the null is true.

A type I error rejects a true null, at rate alpha; a type II error fails to reject a false null, at rate beta. Power is 1 - beta and increases with sample size and effect size. Testing many hypotheses inflates the false positive rate and demands correction.

5. Regression

Simple linear regression fits a line by minimising the sum of squared residuals. The coefficient of determination reports the proportion of variance explained, and says nothing about whether the model is appropriate.

Inference on the coefficients assumes linearity, independent errors, constant variance and approximate normality of residuals. Plotting the residuals is the fastest way to discover that one of these has failed.